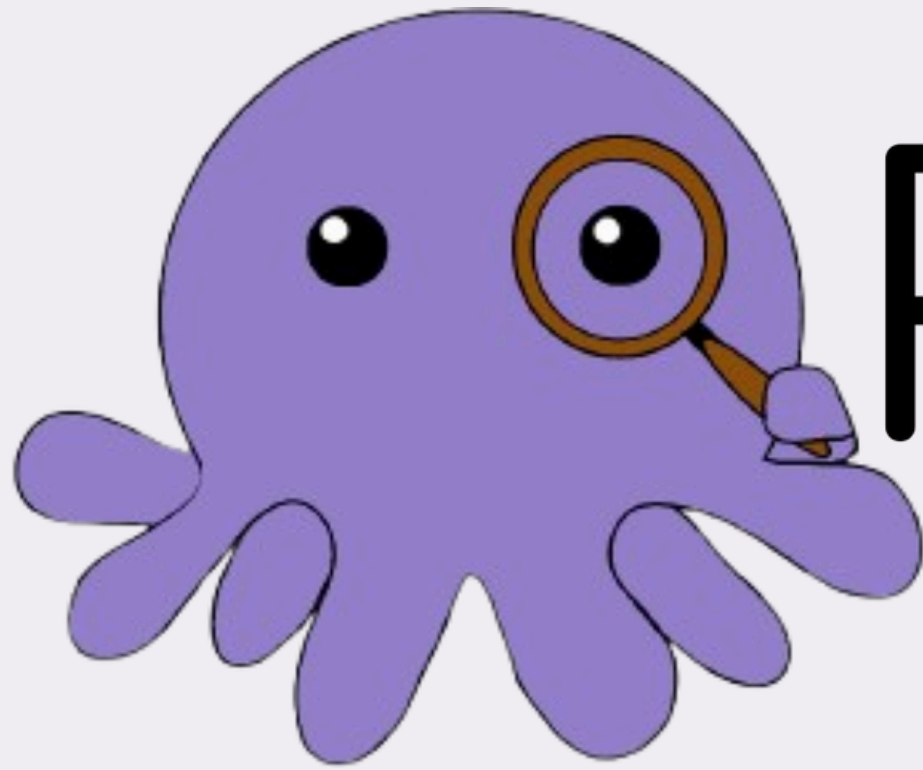




RepoPrint

From Code to Threats, Instantly

Group 3

Artur Yavorskyy - 125275

Gil Guedes - 125031

Margarida Cardoso - 125799

Nuno Costa - 125120

Tiago Costa - 125943

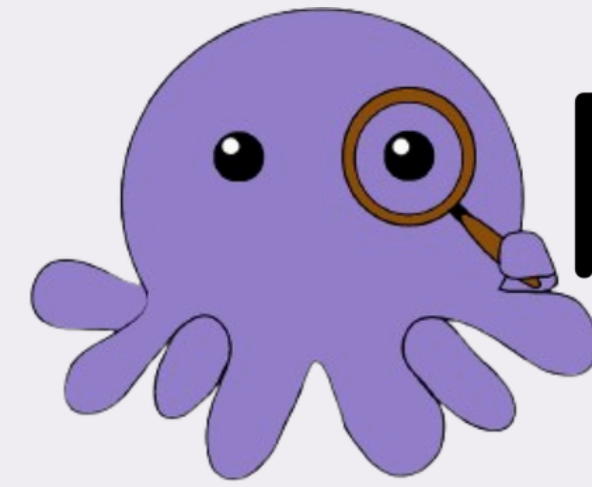
Advisors

João Almeida

Raquel Paradinha

José Gameiro

29/09/2026



RepoPrint

**Automating architectural
security directly from your
Git workflow.**

CONTEXT

- A Git repository has more information than just code
- Information is highly scattered
- Security and Privacy analysis is done manually

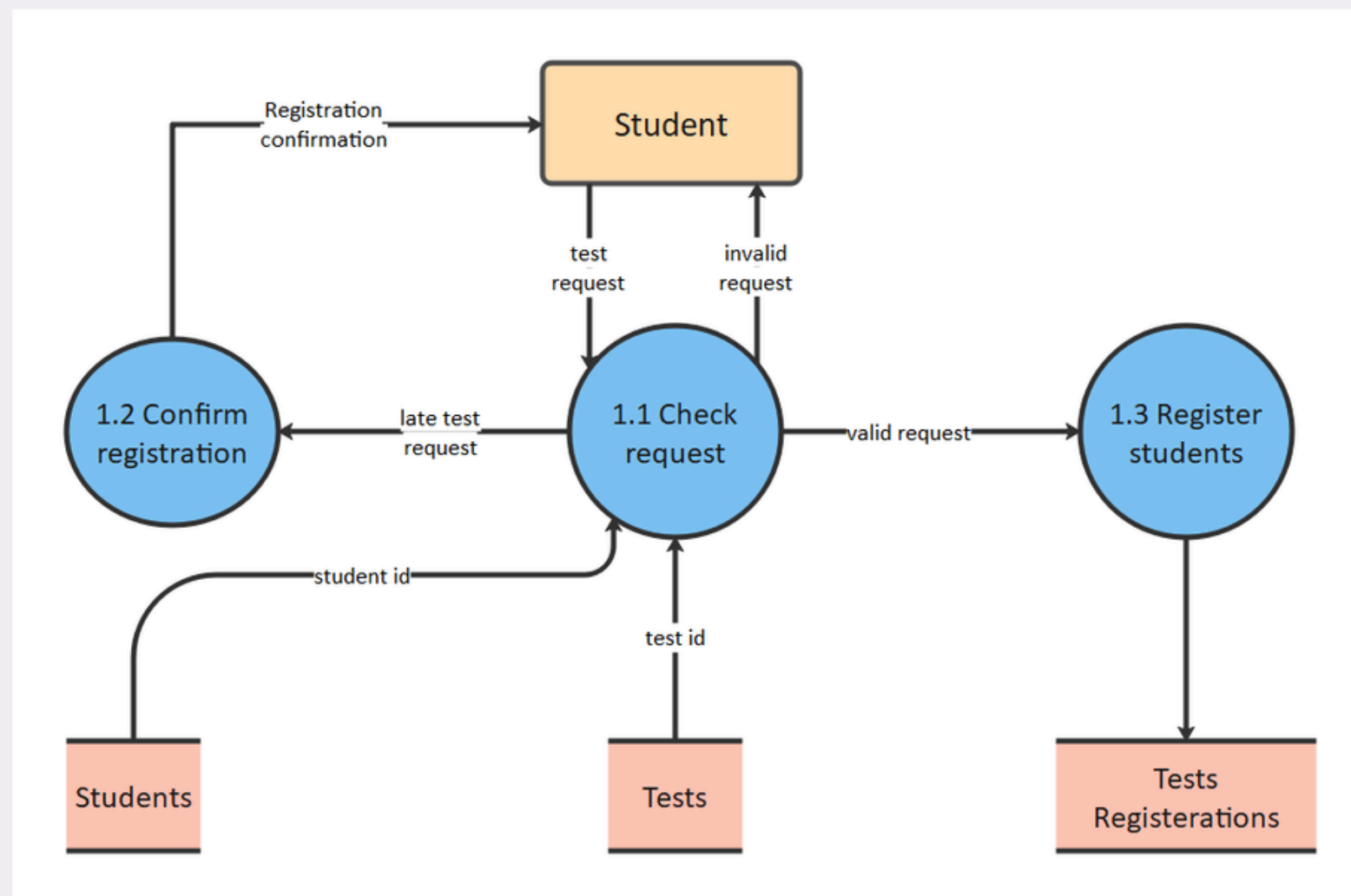


Figure1 - DFD Example

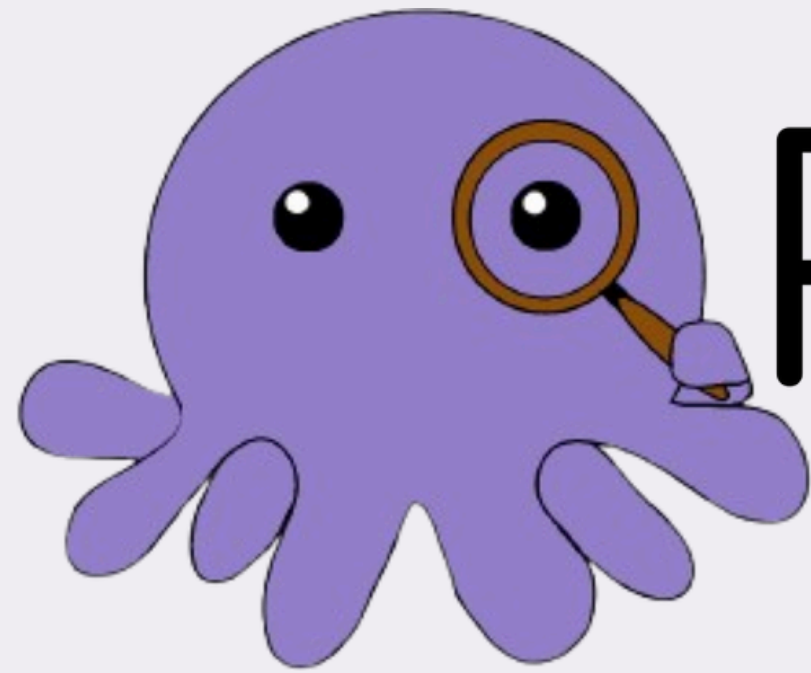
- **STRIDE** - threat modelling framework for cybersecurity
- **LINDDUN** - threat modelling framework for data privacy
- **NIS2, IEC/ISO 27001, RGPD** - norms and laws for security and data privacy
- **DFD** - Data Flow Diagram

PROBLEMS

1. Lack of compliance of security and privacy norms
2. Abundance of poor quality, AI written code
3. Rise in cyberattacks
4. Outdated security systems

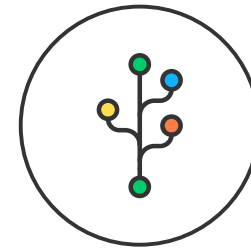


SOLUTION



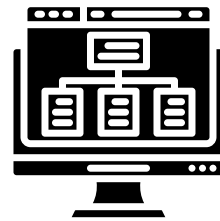
RepoPrint

GOALS



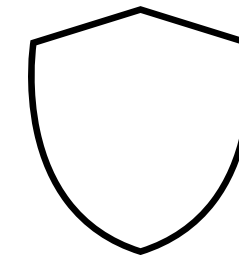
Git Activity & Team Profiling

Commit dynamics, pull request hygiene,
developer contributions



Semi-automated DFD Construction

Static code and IaC analysis to
infer data flows



Threat & Privacy Modeling

STRIDE and LINDDUN compliance

STATE OF THE ART

	GitHub Advanced Security	OWASP Threat Dragon	Sonar	PullRank	RepoPrint
Code & Hygiene Quality	✓	✗	✓	✗	✓
Git & Team Dynamics	✗	✗	✗	✓	✓
Threat & Privacy Modeling	✗	⚠	✗	✗	✓
Auto Generated DFD	✗	✗	✗	✗	✓



Does it, but manually.

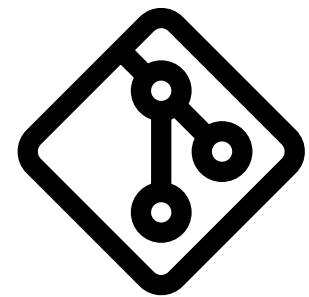


Yes

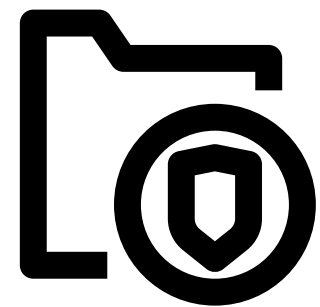


No

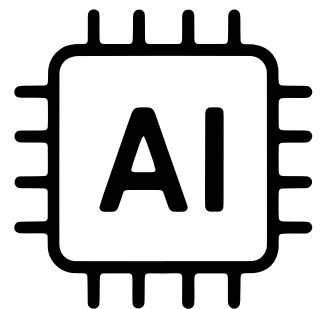
EXPECTED RESULTS



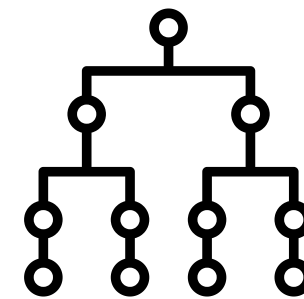
Git Statistics



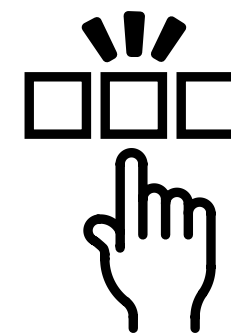
Repo Structure &
Health



SLM Support

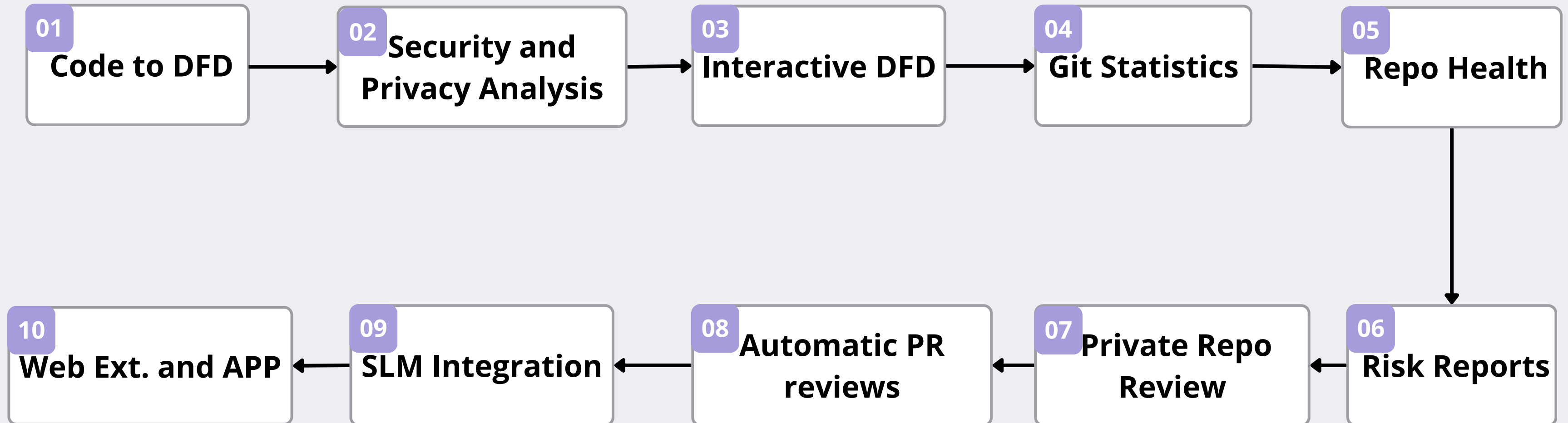


Semi Automatic DFDs

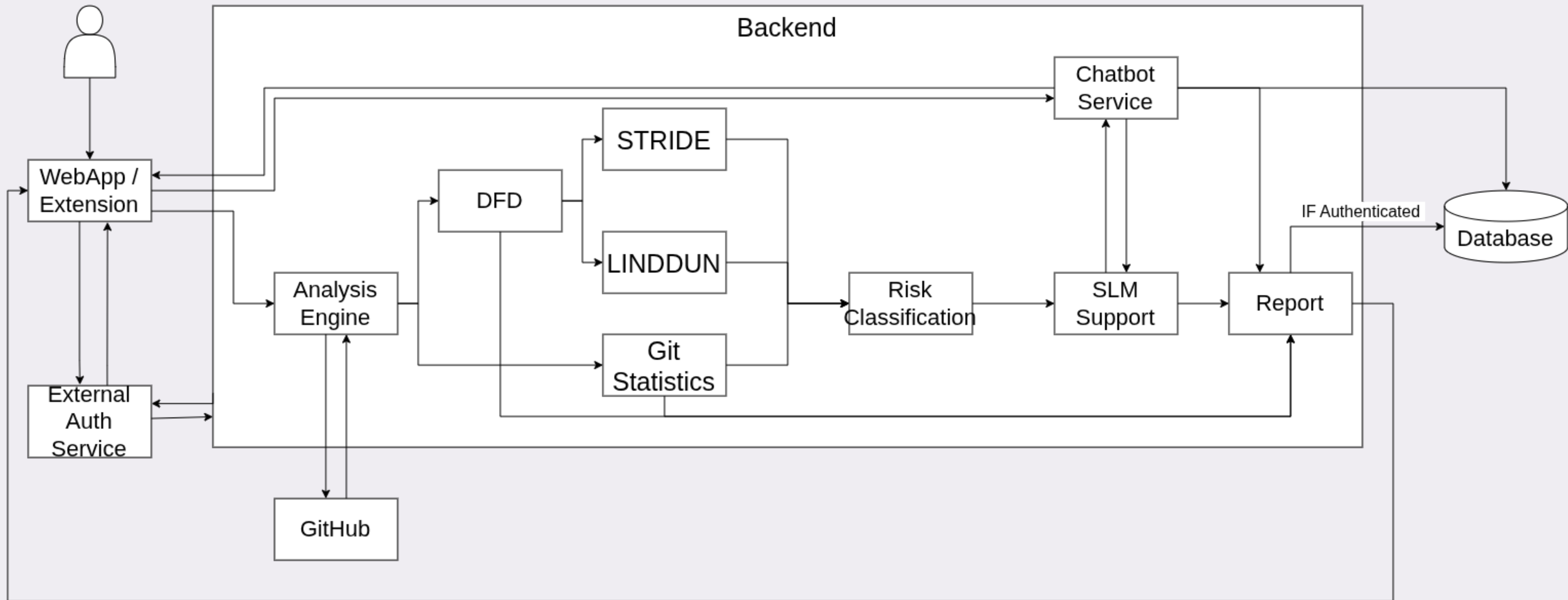


Interactive DFDs

TASKS



HIGH LEVEL ARCHITECTURE



CALENDAR - MILESTONES

Milestone	Date	Deliverables
M1 - Inception	29 September	Presentation, calendar, communication plan
M2 - Elaboration	20 October	Requirements, use cases, architecture, data model
M3 - Construction	2 November	Core API, database, frontend skeleton, DFD prototype
M4 - Construction	16 December	MVP, report, audit results

Open Up	Weekly focus	Descriptions	Deliverables	Designated Dev	Start Date	End Date	Status
Inception	 Milestone 1 Preparation		• GitHub configuration • Microsite • Calendar • Logo • Presentation	Nuno Margarida Tiago Gil Artur	September 22, 2026	September 28, 2026	Done
Presentation	 Presentation Rehearsel			Everyone	September 28, 2026	September 29, 2026	Done
Inception	 Milestone 1	Focuses on defining the project foundation and validating the concept prior to moving into in-depth development.	• Project presentation • Project calendar • Communication plan	Everyone	September 29, 2026	September 29, 2026	In progress
Elaborarion	 Existing technologies and State of the Art				September 30, 2026	October 7, 2026	Not started
Elaborarion	 Use cases, Personas, Scenarios				September 30, 2026	October 7, 2026	Not started
Elaborarion	 System Architecture				September 30, 2026	October 12, 2026	Not started
Elaborarion	 Functional and non Functional requirements				September 30, 2026	October 12, 2026	Not started
Elaborarion	 DER and ER of the Database				September 30, 2026	October 12, 2026	Not started
Presentation	 Presentation Planning				October 11, 2026	October 13, 2026	Not started
Elaborarion	 Milestone 2	The team must analyze requirements, identify actors and use cases, define technical and non-functional constraints, and validate the proposed solution. It also aims to mitigate risks, improve effort estimation, and establish a stable architectural baseline for the next phase.	• Requirements gathering • Functional requirements • Context and State of The Art • Actors • Use Cases • Non-functional requirements • Deployment diagram • Domain model [?] • Mock-ups [?]	Everyone	October 13, 2026	October 20, 2026	Not started

📁 Open Up	➕ Weekly focus	≡ Descriptions	≡ Deliverables	⋮ Designated Dev	📅 Start Date	📅 End Date	⚙ Status
Construction	📄 Database Construction				October 14, 2026	November 2, 2026	● Not started
Construction	📄 Frontend skeleton				October 14, 2026	November 2, 2026	● Not started
Construction	📄 Build main EndPoints				October 14, 2026	November 2, 2026	● Not started
Construction	📄 Construction of a code to DFD tool (may take more time)				October 14, 2026	November 2, 2026	● Not started
Construction	📄 Legal Requirements Compliance and Protection		• RGPD • Licensing and Intellectual Property • Industry-Specific Regulation • Terms and Conditions		October 14, 2026	November 2, 2026	● Not started
Construction	📄 Technical Risks Analysis and Mitigation		• Interoperability and External Dependencies • Technical Debt and Maintainability • Fragility of Critical Components		October 14, 2026	November 2, 2026	● Not started
Construction	📄 Business Plan Strategic Analysis		• SWOT Analysis • TOWS Matrix • PESTEL Analysis		October 14, 2026	November 2, 2026	● Not started
Presentation	📄 Presentation Planning				November 2, 2026	November 3, 2026	● Not started
Construction	📅 Milestone 3	Identify relevant legal requirements, assess technical risks, and understand the external environment in which the solution will operate.	• Legal Requirements Compliance and Protection • Technical Risks Analysis and Mitigation • Business Plan Strategic Analysis	Everyone	November 3, 2026	December 14, 2026	● Not started

Open Up	Weekly focus	Descriptions	Deliverables	Designated Dev	Start Date	End Date	Status
		the solution will operate.	• Business Plan Strategic Analysis				
Construction	Interactive DFD				November 4, 2026	November 24, 2026	Not started
Construction	Git Statistics				November 4, 2026	November 24, 2026	Not started
Construction	Performance		- Response Time Metrics - Throughput Metrics - Identification of Slow Database Queries		November 4, 2026	December 14, 2026	Not started
Construction	Scalability		- Stress Testing - Architectural Proof of Concept		November 4, 2026	December 14, 2026	Not started
Construction	Security	Demonstrate that security concerns existed. Explain how we thought about security.			November 4, 2026	December 14, 2026	Not started
Construction	AI Model and Algorithm Validation	- Ability to handle unseen data. - Perform error analysis - Induce errors in critical system components			November 4, 2026	December 14, 2026	Not started
Construction	Software Project Management	Evaluation and justification of: - Project agile planning - Automation and CI/CD - Version control and PR reviews			November 4, 2026	December 14, 2026	Not started
Construction	CheckPoint			Everyone	November 24, 2026	November 24, 2026	Not started
Construction	Repo Health				November 25, 2026	December 14, 2026	Not started
Construction	Risk Reports				November 25, 2026	December 14, 2026	Not started
Construction	Deploy in VMs				November 25, 2026	December 14, 2026	Not started
Construction	MVP				November 25, 2026	December 14, 2026	Not started
Presentation	Presentation Planning				December 11, 2026	December 14, 2026	Not started
Construction	Milestone 4	The objective of this milestone is to evaluate the strategic and economic feasibility of the project, consolidating the business vision and competitive landscape.	- Performance - Scalability - Security - AI Model and Algorithm Validation - Software Project Management - MVP	Everyone	December 15, 2026	December 16, 2026	Not started

	📄 Improvement of the MVP				February 8, 2027	March 23, 2027	● Not started
	📄 Private Repository Reviews				February 8, 2027	March 23, 2027	● Not started
	📄 Automatic PR Reviews				February 8, 2027	March 23, 2027	● Not started
	📄 SLM Integration				February 8, 2027	March 23, 2027	● Not started
	📄 Extension Construction				February 8, 2027	March 23, 2027	● Not started
	📄 User Testing				April 1, 2027	April 8, 2027	● Not started
	📄 Apply Users' ideas				April 8, 2027	April 21, 2027	● Not started
	📄 Stabilize final product				April 21, 2027	May 18, 2027	● Not started
	📄 Write Documentation				May 18, 2027	May 28, 2027	● Not started
Presentation	📄 Plan Presentation				May 25, 2027	June 1, 2027	● Not started
	📄 Students@DETI		• Demo • Poster • Video • Technical report	Everyone	June 2, 2027	June 3, 2027	● Not started

OUR ROLES



Artur Yavorsky
Product Owner



Gil Guedes
Quality Assurance



Margarida Cardoso
Team Manager



Nuno Costa
Architect



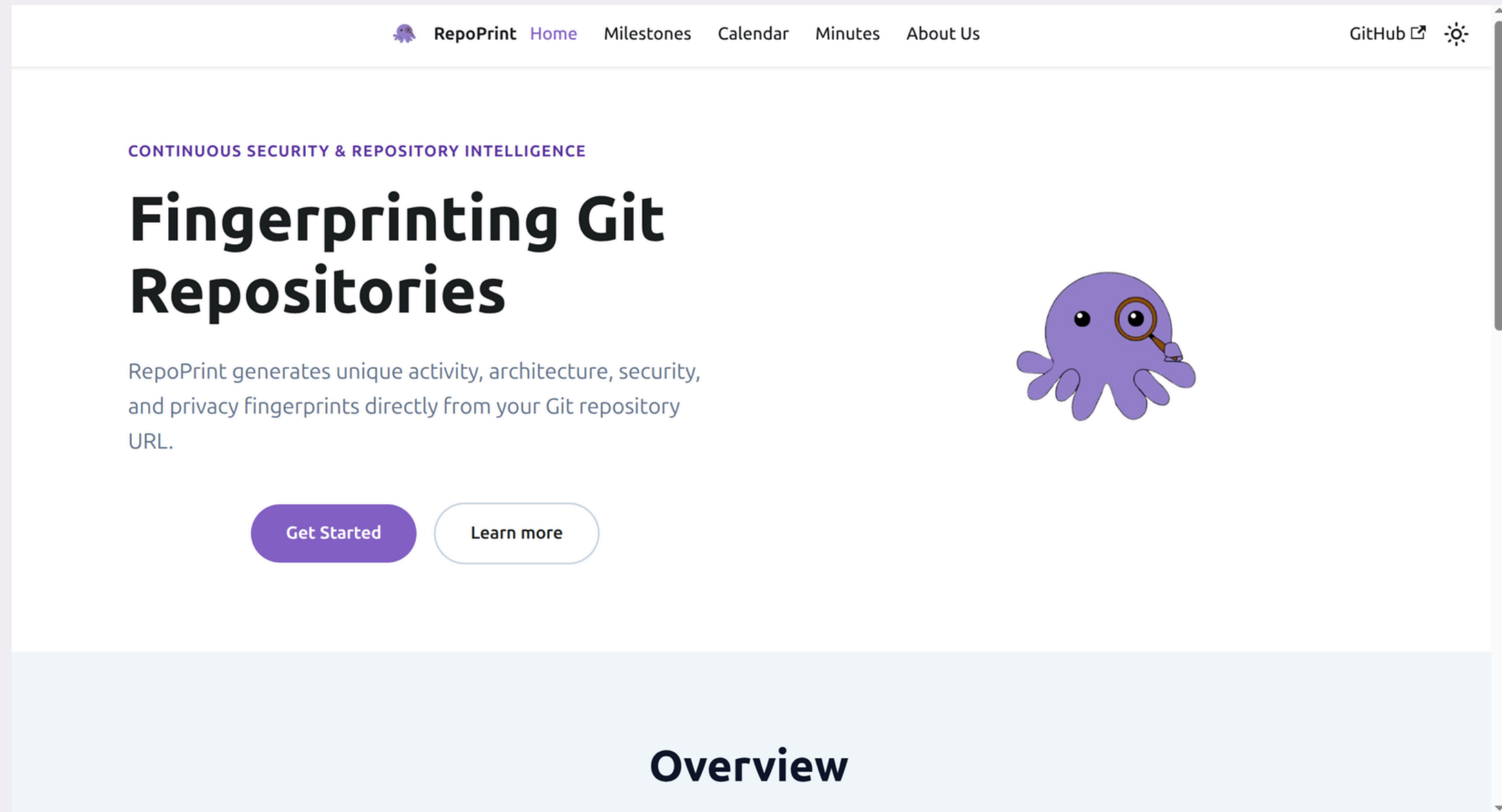
Tiago Costa
DevOps

COMMUNICATION AND ASSIGNMENTS





<https://gitrepoprint.github.io/Microsite/>



Keep up with our work!